

MXM Open House

Tuesday, December 9

Advertisement

- **Date:** December 9
- **Time:** 2-5 pm
- **Location:** Van Vleck 911
- Poster session to see results of undergraduate research this semester!
- Coffee & cookies

Math 421

Monday, December 8

Announcements

- This week's drop-in hours –
 - Wednesday 12:00-1:00 pm VV B205
 - Thursday 9:00 – 11:00 am VV B205
- Final Exam –
 - Date: Monday, December 15th
 - Time: 5:05-7:05 pm
 - Location: Bardeen 140
- Course Evaluations!
 - Open until 11:59pm on 12/10

Equivalent Definition

Theorem 32.5. (“Cauchy criterion for integrability”) If f is bounded on $[a, b]$, then f is integrable on $[a, b]$ if and only if

for every $\epsilon > 0$, there exists a partition P of $[a, b]$

such that $U(f, P) - L(f, P) < \epsilon$

Example

Let $f(x)$ be defined as follows. Prove f is integrable on $[-1,1]$.

$$f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$$

Monotonicity $\Rightarrow$ Integrability

Theorem 33.1. Every monotonic function f on $[a, b]$ is integrable.

Proof: Assume f increasing.

1. Why is f bounded?
2. What does $U(f, P) - L(f, P)$ look like?
3. What restriction should you impose on the length of each subinterval so that $U(f, P) - L(f, P) < \epsilon$?

Continuity $\Rightarrow$ Integrability

Theorem 33.2. Every continuous function f on $[a, b]$ is integrable.

Proof Ingredients:

Continuity $\Rightarrow$ Integrability

Discussion Questions

- 1 Consider the set of all real-valued functions f defined on the closed interval $[a, b]$. Describe the relationship between the subsets of continuous, differentiable, and integrable functions.
- 2 How discontinuous can a function be while still being integrable?

IVT for integrals

Thm. If f is a continuous function on $[a, b]$, then for at least one $x \in (a, b)$, we have

$$f(x) = \frac{1}{b-a} \int_a^b f(x) dx$$

Activity – Prove the Theorem

Step 1: Let M and m be the max/min of f on $[a, b]$. Assume $m < M$. What happens if $m = M$?

Step 2: Convince yourself that

$$m(b - a) \leq \int_a^b f \leq M(b - a)$$

This fact holds for any integrable f !

Step 3: Use the IVT to get the desired result.

Some Properties of Integration

- If f is integrable on $[a, b]$, then for any number c , the function cf is integrable on $[a, b]$ and $\int_a^b cf = c \cdot \int_a^b f$.
- If f and g are integrable on $[a, b]$, then $f + g$ is integrable on $[a, b]$ and $\int_a^b (f + g) = \int_a^b f + \int_a^b g$.
- If f and g are integrable on $[a, b]$ and if $f(x) \leq g(x)$ for $x \in [a, b]$, then $\int_a^b f \leq \int_a^b g$.

More Properties of Integration

- If g is a continuous nonnegative function on $[a, b]$ and if $\int_a^b g = 0$, then g is identically 0 on $[a, b]$.
- If f is integrable on $[a, b]$, then $|f|$ is integrable on $[a, b]$ and

$$\left| \int_a^b f \right| \leq \int_a^b |f|$$

- If $a < c < b$ and f is integrable on $[a, b]$, then

$$\int_a^b f = \int_a^c f + \int_c^b f$$

Equivalent Definitions

Definition. A function f which is bounded on $[a, b]$ is **integrable** on $[a, b]$ if

$$L(f) = U(f)$$

Theorem 32.5. (“Cauchy criterion for integrability”) If f is bounded on $[a, b]$, then f is integrable on $[a, b]$ if and only if

for every $\epsilon > 0$, there exists a partition P of $[a, b]$

such that $U(f, P) - L(f, P) < \epsilon$

Proof of Equivalence ($\Rightarrow$)

Proof of Equivalence ($\Leftrightarrow$)